

Question: Define estimation, point estimation; What are the criteria of a good point estimator? Explain them.

Answer:

✓ Estimation: Let X be a random variable whose density is $f(x, \theta)$, where the form of the density is assumed known with parameter θ . Further assume that the values $x_1, x_2, \dots, x_n$ of a random sample $X_1, X_2, \dots, X_n$ from $f(x, \theta)$ can be observed. On the basis of the observed sample values $x_1, x_2, \dots, x_n$, it is desired to estimate the value of the unknown parameter θ , or the value of some function, say $T(\theta)$, of the parameter and the process is known as estimation.

✗ Point estimation: The point estimation is to let the value of some statistic, say $t(x_1, x_2, \dots, x_n)$ represent or estimate, the unknown $T(\theta)$, where $T(\theta)$ is a function of the parameter θ , such a statistic $t(x_1, x_2, \dots, x_n)$ is called a point estimation.

The criteria of a good point estimator are given below —

- a) Unbiasedness
- b) Consistency
- c) Sufficiency and
- d) Efficiency or minimum variance.

a) Unbiasedness: Any statistic whose mathematical expectation is equal to parameter θ is called an unbiased estimator of the parameter θ . Otherwise, the statistic is called biased.

Thus for unbiasedness of an estimator $\hat{\theta}$ of the parameter θ , we must have $E(\hat{\theta}) = \theta \forall \theta$.

b) Consistency: An estimator $\hat{\theta}_n$ computed from random sample of n observations is said to be consistent for population parameter θ , if it converges to θ as $n \rightarrow \infty$, i.e. for given two positive number ϵ and δ however small, we can find a positive integer N depending on ϵ and δ such that

$$P[|\hat{\theta}_n - \theta| < \epsilon] > 1 - \delta ; \forall n > N$$

$$\text{or, } P[\lim_{n \rightarrow \infty} \hat{\theta}_n = \theta] = 1.$$

c) Sufficiency: An estimator is said to be sufficient for a parameter if it contains all the information in the sample regarding the parameter.

d) Efficiency: Efficiency is an inverse function of variance of an estimator. Let us suppose for large sample, two consistent estimators T_1 and T_2 both of them are distributed asymptotically normally about true value of the parameter θ with variance $\frac{\sigma_1^2}{n_1}$ and $\frac{\sigma_2^2}{n_2}$ respectively. Then T_1 is said to be more efficient than T_2 if $\sigma_1^2 < \sigma_2^2$. However if $v(T_1) < v(T_2)$ for all n . To remember that the efficiency of an estimator is denoted by e and is defined as

$$e(\hat{\theta}) = \frac{\text{Variance of the most efficient estimator}}{\text{Variance of the given estimator}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} e(\hat{\theta}) = e_0(\hat{\theta})$$

Obviously $0 \leq e_0(\hat{\theta}) \leq 1$, when $e_0(\hat{\theta}) = 1$ then $\hat{\theta}$ is called an asymptotically efficient estimator of θ .

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Problem: Let $x_1, x_2, \dots, x_n$ be the random sample from a normal density with mean μ and variance σ^2 . Show that $\bar{x}$ is an unbiased estimator of μ , but $S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is not an unbiased estimator of σ^2 . Also show that $\bar{x}$ is a consistent estimator of μ .

Proof: Let $x_1, x_2, \dots, x_n$ be the r.v. from $N(\mu, \sigma^2)$. The sample ^{estimator} mean of μ is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \hat{\mu}$.

By the definition of unbiasedness we have

$$\begin{aligned} E(\hat{\mu}) &= E(\bar{x}) = E\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n} \sum_{i=1}^n E(x_i) \\ &= \frac{1}{n} [E(x_1) + E(x_2) + \dots + E(x_n)] \\ &= \frac{1}{n} [\mu + \mu + \dots + \mu] \\ &= \mu \end{aligned}$$

$$\Rightarrow E(\hat{\mu}) = \mu$$

Hence $\bar{x} = \hat{\mu}$ is an unbiased estimator of μ .

Now

$$\begin{aligned} \sum_{i=1}^n (x_i - \mu)^2 &= \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \mu)^2 \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n (\bar{x} - \mu)^2 + 2(\bar{x} - \mu) \sum_{i=1}^n (x_i - \bar{x}) \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \quad \because \sum_{i=1}^n (x_i - \bar{x}) = 0 \end{aligned}$$

$$\Rightarrow \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2$$

$$\begin{aligned} \therefore E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right] &= E\left[\sum_{i=1}^n (x_i - \mu)^2\right] - nE(\bar{x} - \mu)^2 \\ &= \sum_{i=1}^n E(x_i - \mu)^2 - nE(\bar{x} - \mu)^2 \\ &= nV(x) - n \frac{V(x)}{n} = (n-1)V(x) \\ &= (n-1)\sigma^2 \end{aligned}$$

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We have sample variance

$$S^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$\therefore E(S^2) = \frac{1}{n} E \left[\sum (x_i - \bar{x})^2 \right] = \frac{n-1}{n} \sigma^2 \neq \sigma^2$$

Hence $S^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ is not an unbiased estimator of σ^2 .

Since $X \sim N(\mu, \sigma^2)$

Then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$,

$$\text{i.e., } V(\bar{X}) = \frac{\sigma^2}{n}$$

$$\text{Then } \lim_{n \rightarrow \infty} V(\bar{X}) = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} \rightarrow 0$$

Hence the sample mean $\bar{x}$ is consistent estimator of μ .

Question: What are the methods of point estimation? Define Likelihood function, Maximum likelihood estimator. Describe the principle of maximum likelihood method of estimation.

Answer:

The methods of point estimation are as follows -

- Method of Least Squares
- Method of Moments
- Method of Minimum Chi-square
- Method of Maximum Likelihood
- Bayes' Method
- Mini-max Method.

Likelihood function: The likelihood function of n random variable $x_1, x_2, \dots, x_n$ is defined to be the joint density of the n random variable say $f(x_1, x_2, \dots, x_n)$ which is considered to be function of the parameter of the distribution.

In particular if $x_1, x_2, \dots, x_n$ be a random sample from the density $f(x; \theta)$ then the likelihood is usually denoted by L is given by

$$L = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) \\ = \prod_{i=1}^n f(x_i; \theta)$$

Maximum likelihood estimator (MLE): Let $L(\theta) = L(\theta; x_1, x_2, \dots, x_n)$ be the likelihood function of the random variable X of the random sample $x_1, x_2, \dots, x_n$. If $\hat{\theta}$ [where $\hat{\theta} = \theta(x_1, x_2, \dots, x_n)$] is a function of the observations $x_1, x_2, \dots, x_n$ is the value of the θ in Ω which maximized $L(\theta)$ then $\hat{\theta} = \theta(x_1, x_2, \dots, x_n)$ is called the maximum likelihood estimator for θ given that the sample $x_1, x_2, \dots, x_n$.

The principle of maximum likelihood method of estimation:

Let us consider a random sample $x_1, x_2, \dots, x_n$ from a distribution having pdf $f(x; \theta)$, $\theta \in \Omega$. The joint pdf of $x_1, x_2, \dots, x_n$ is $f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)$. This joint pdf may be regarded as a function of θ . When so regarded it is called the likelihood function L of the r.v. and we may write

$$L(\theta; x_1, x_2, \dots, x_n) = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) \\ \text{or, simply, } L(\theta | X) = \prod_{i=1}^n f(x_i; \theta)$$

The principle of maximum likelihood consists in finding an estimator of the parameter which maximizes L in the parameter.

Suppose that we can find a function of $x_1, x_2, \dots, x_n$, say $u(x_1, x_2, \dots, x_n)$, such that when θ is replaced by $u(x_1, x_2, \dots, x_n)$ the likelihood function L is maximum, i.e., $L[u(x_1, x_2, \dots, x_n); x_1, x_2, \dots, x_n]$ is at least as great as $L(\theta; x_1, x_2, \dots, x_n)$ for every $\theta \in \Omega$. Then the statistic $u(x_1, x_2, \dots, x_n)$ will be called a maximum likelihood estimator (MLE) of θ and will be denoted by the symbol $\hat{\theta} = u(x_1, x_2, \dots, x_n)$.

Thus $\hat{\theta}$ is the solution, if any of

$$\frac{\partial L}{\partial \theta} = 0 \quad \text{and} \quad \frac{\partial^2 L}{\partial \theta^2} < 0;$$

Since $L > 0$, so $\log L$ which shows that L and $\log L$ attain their extreme values at the same value of $\hat{\theta}$. Then we can write

$$\frac{\partial L}{\partial \theta} = \frac{\partial \log L}{\partial \theta} = 0$$

and solving the equation we shall get the MLE of θ .

Problem: Let $x_1, x_2, \dots, x_m$ be a r.v. from $f(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$; $x = 0, 1, 2, \dots, n$. Find the MLE of p .

Solution: The likelihood function L is given by

$$\begin{aligned} L &= \prod_{i=1}^m f(x_i; p) \\ &= \left\{ \prod_{i=1}^m \binom{n}{x_i} \right\} p^{\sum x_i} (1-p)^{mn - \sum x_i} \\ \Rightarrow \log L &= \log \left\{ \prod_{i=1}^m \binom{n}{x_i} \right\} + \left(\sum x_i \right) \log p + (mn - \sum x_i) \log (1-p) \\ \Rightarrow \frac{\partial \log L}{\partial p} &= 0 + \frac{\sum x_i}{p} - \frac{mn - \sum x_i}{1-p} \end{aligned}$$

$$\Rightarrow 0 =$$

$$\Rightarrow \frac{\sum_{i=1}^m x_i}{\hat{p}} - \frac{mn - \sum_{i=1}^m x_i}{1 - \hat{p}} = 0$$

$$\Rightarrow \sum_{i=1}^m x_i - \hat{p} \sum_{i=1}^m x_i - \hat{p} mn + \hat{p} \sum_{i=1}^m x_i = 0$$

$$\Rightarrow \hat{p} mn = \sum_{i=1}^m x_i$$

$$\Rightarrow \hat{p} = \frac{1}{mn} \sum_{i=1}^m x_i = \frac{\bar{x}}{n}$$

$$\because \bar{x} = \frac{1}{m} \sum_{i=1}^m x_i$$

$$\Rightarrow \hat{p} = \frac{\bar{x}}{n} \text{ is a MLE of } p.$$

Problem: Let $x_1, x_2, \dots, x_n$ be a r.v. from $f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}; x=0, 1, 2, \dots, \infty$.

Find MLE of λ .

Solution: The likelihood function is given by

$$L = \prod_{i=1}^n f(x_i; \lambda)$$

$$= \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= \frac{e^{-\lambda n} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

$$\Rightarrow \log L = -\lambda n + \left(\sum_{i=1}^n x_i \right) \log \lambda - \log \left(\prod_{i=1}^n x_i! \right)$$

$$\Rightarrow \frac{\partial \log L}{\partial \lambda} = -n + \frac{\sum_{i=1}^n x_i}{\lambda}$$

$$\Rightarrow -n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0$$

$$\Rightarrow n = \frac{1}{\lambda} \sum_{i=1}^n x_i$$

$$\Rightarrow \hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

Hence $\hat{\lambda} = \bar{x}$ is a MLE of λ .

KE-239
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2018
Problem: Let $(x_1, x_2, \dots, x_n)$ be a random sample from a normal population $N(\mu, \sigma^2)$. Find the MLE of μ & σ^2 .

Let $x_1, x_2, \dots, x_n$ be a r.s. from $f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$, $-\infty < x < \infty$, $-\infty < \mu < \infty$, $\sigma > 0$. Find the MLE of μ and σ^2 .

Solution: The pdf of normal distribution is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The likelihood function L is given by

$$L = \left(\frac{1}{\sqrt{2\pi}}\right)^n \frac{1}{(\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}$$

$$\Rightarrow \log L = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\Rightarrow \left\{ \begin{array}{l} \frac{\partial \log L}{\partial \mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) \\ \frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 \end{array} \right.$$

$$\Rightarrow \frac{1}{\hat{\sigma}^2} \sum (x_i - \hat{\mu}) = 0 \quad \text{and} \quad -\frac{n}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4} \sum (x_i - \hat{\mu})^2 = 0$$

$$\Rightarrow \sum x_i = n\hat{\mu} \quad \text{and} \quad -n\hat{\sigma}^2 + \sum (x_i - \hat{\mu})^2 = 0$$

$$\Rightarrow \hat{\mu} = \frac{1}{n} \sum x_i = \bar{x} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \hat{\mu})^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = s^2$$

Therefore $\bar{x} = \frac{1}{n} \sum x_i$ and $s^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ are the MLE of μ and σ^2 respectively.